



NGEE ANN
P O L Y T E C H N I C

SAMPLE FINAL TEST

School of Engineering
(Diploma in Aerospace Engineering)

Level 2

Time Allowed: 1.5 Hours

AIRCRAFT STRUCTURES & MATERIALS
(012648)

INSTRUCTIONS TO CANDIDATES:

1. **Check carefully to ensure you are sitting for the correct paper.**
2. Answer All questions.

SECTION A – Multiple-Choice Questions (15 marks)

SECTION B – Short-Answer Questions (45 marks)

SECTION C – Analytical Questions (40 marks)

3. If more than the required number of questions is attempted, only the required number of answers in the order in which the questions were attempted in the Answer Book shall be marked.
4. Begin each question on a separate page.
5. This paper consists of 13 pages including this cover page. Check carefully to make sure your set is complete.

SECTION A – Multiple-Choice Questions (Total marks: 15 marks, 1 mark each)

Tick (✓) the correct answer on the answer sheet provided on page 1 of the Answer Book.

Question 1

1.1 Which one of the statements below is correct?

- (A) Window openings are centered between the fuselage stringers.
- (B) Window openings are centered between fuselage sections.
- (C) Window openings are centered between the fuselage frames.
- (D) Window openings are centered between the fuselage shear ties.

C

1.2 Which of the following is NOT a member of a semimonocoque structure?

- (A) Stringers
- (B) Ribs
- (C) Bulkheads
- (D) Frames

B

1.3 In general, airplanes are designed to withstand _____ times the maximum expected forces.

- (A) 1.25
- (B) 1.5
- (C) 1.75
- (D) 2

B

1.4 Which one of the principal load-carrying member is subjected to the greatest load during flight?

- (A) Wings
- (B) Fuselage
- (C) Landing Gears
- (D) Vertical Stabilisers

A

1.5 Tabs are secondary flight control surfaces set into the _____ of the primary surfaces.

- (A) trailing edges
- (B) leading edges
- (C) outboard edges
- (D) inboard edges

A

1.6 What is the streamline fairing installed between vertical stabiliser and fuselage?

- (A) Rudder guides
- (B) Vertical stabiliser to body fairing
- (C) Dorsal fin
- (D) Ventral fin

C

1.7 Which of the following groups are considered to be secondary flight controls?

- (A) Tabs, ailerons, spoilers, and slats.
- (B) Tabs, flaps, elevators, and slats.
- (C) Rudder, flaps, spoilers, and slats.
- (D) Tabs, flaps, spoilers, and slats.

D

1.8 Major zone 300 refers to the _____.

- (A) Lower half of fuselage
- (B) Empennage
- (C) Powerplants and pylons
- (D) Doors

B

1.9 Wing spars is the principal _____ member of the wing structure.

- (A) chordwise
- (B) spanwise
- (C) longitudinal
- (D) circumferential

B

- 1.10 What is the condition associated with the alignment of the wheels shown in Figure 1.10 known as?

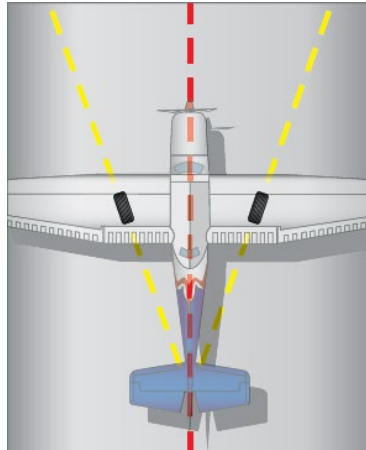


Figure 1.10

- (a) Positive camber
- (b) Negative camber
- (c) Toe-in
- (d) Toe-out

D

- 1.11 What does the letter “R” refer to in the 3-part size designation marking for a tyre (e.g. 34X9.25R16)?

- (a) Nominal outer diameter
- (b) Nominal section width
- (c) Radial tyre
- (d) Rim diameter

C

- 1.12 The red dot on side of tire position represent the _____ point of the tyre.

- (a) thickest
- (b) thinnest
- (c) heaviest
- (d) lightest

D

- 1.13 Fairings are a type of _____ structures.

- (a) primary
- (b) secondary
- (c) tertiary
- (d) load-carrying

C

1.14 If the horizontal stabiliser is attached to the fuselage with a pivoting hinge as its rear spar, where will the actuator for the horizontal stabiliser be located?

- (A) Rear spar of horizontal stabiliser.
- (B) Front spar of horizontal stabiliser.
- (C) Auxiliary spar of horizontal stabiliser.
- (D) False spar of horizontal stabiliser

B

1.15 Which of the following is NOT a type of aircraft station?

- (A) Fuselage station
- (B) Butt line
- (C) Water line
- (D) Aircraft zones

D

SECTION B – Short-Answer Questions (Total marks: 45 marks)

Question 2 (15 marks)

2.1 Define the term “load factor”.

(2 marks)

Load factor is the ratio of the total load supported by the airplane's wing to the actual weight of the airplane and its contents by the wings divided by the total weight of the airplane.

2.2 Define the term “ultimate load”.

(2 marks)

Ultimate load is 1.5 times the limit load factor and the limit load factor is the maximum of the normally anticipated loads the aircraft structure must withstand. The structure must be able to support ultimate loads without failure for at least 3 seconds.

2.3 Study the fuselage structure in Figure 2.3.

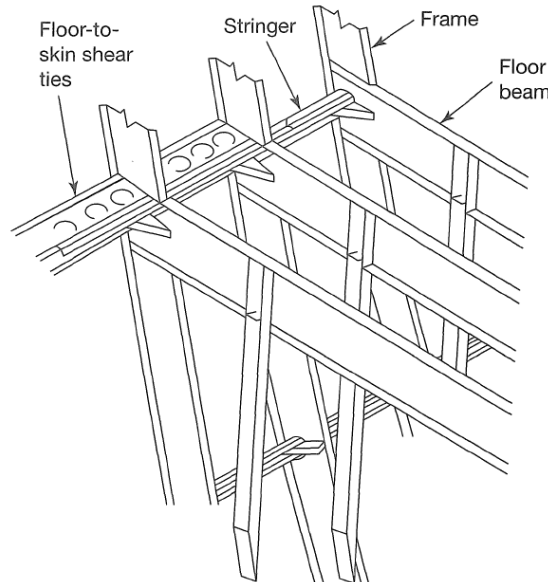


FIGURE 2-23 Structural floor supports. (Boeing Commercial Aircraft Co.)

Figure 2.3

a.) Describe the component “shear tie” and list its functions.

(2 marks)

Shear ties extend longitudinally along the intersection of the floor beams and

frame.

They help to transfer payloads/dead weight from the floor beams to the skin and frames.

b.) Describe the component “floor beam” and list its functions.

(2 marks)

Floor beams provide support to cabin floor, which payloads rest upon.

Floor beams are attach to the frames and run horizontally across the fuselage.

2.4 Briefly explain why keel beams are only located at the fuselage belly.

(2 marks)

Keel beam help to reinforce the wing center section and wheel areas due to openings and fittings

OR

During flight, much compression loads take place on the fuselage belly and the keel beam is there to reinforce the fuselage against buckling/bending.

2.5 Describe how a ruddervator can serve the function of the rudder and elevators. You may wish to use a sketch to supplement your explanation.

(3 marks)

Ruddervator’s surfaces are mounted at an angle above horizontal.

When serving as elevators, the surfaces on each side of the tail move in the same direction, either up or down.

When serving as a rudder, the surfaces move in opposite directions, one up and one down.

State any TWO types of primary flight control tabs commonly found on airplanes.

(2 marks)

Any 2 of the following:

1. Ailerons
2. Elevators
3. Rudders

Question 3 (15 marks)

3.1 What is the primary structural design difference between a cantilever wing and a semicantilever wing?

(2 marks)

A cantilever wing contains all its structural strength inside the wing structure and requires no external bracing while a semicantilever wing obtains its strength both by internal wing design and external support and by bracing from struts and wires.

3.2 Define the term “integral fuel tank”

(3 marks)

An integral fuel tank is a tank that is part of the basic structure of the aircraft and cannot be removed because they are an integral and permanent part of the aircraft structure.
Integral fuel tanks have commonly been located in the wing, fuselage or horizontal stabilizer.

3.3 State any THREE fuel tank components which are typically found on all types fuel tanks.

(3 marks)

Any of the three below:

1. Fuel sumps / drains
2. Fuel pumps
3. Vents
4. Baffles

3.4 State any TWO fuel tank requirements that governed fire risk.

(4 marks)

- Fuel tanks must not be installed on the engine side of the fire wall or in any fire zone unless construction and arrangement can prove that it would be safe as if the tanks were not in the zone.

- At least 0.5 inch (12.7mm) of clear air space must be provided between the fuel tank and firewall and between the tank and any fire zone.

Or other relevant requirements.

3.5 Briefly explain why it is important to electrostatic bonding the access door of integral fuel tank.

(3 marks)

Electrostatic bonding prevents the build-up of static charges during flight and prevents the possibility of sparks, which can be ignited the fuel in the integral fuel tank.

Or

Electrostatic bonding prevents the possibility of lightning strikes entering the fuel tanks, which can be ignited the fuel in the integral fuel tank, by allowing rapid discharge of the electrical charges.

Question 4 (15 marks)

4.1. Describe what non-absorbing landing gear is.

(2 marks)

Non-absorbing LDGs DO NOT dissipate the energy of the aircraft contacting the ground during landing.

It only temporary store the energy and quickly return it to the aircraft in a form that is non-damaging to the airframe, and/or passengers.

4.2. Describe what a centering cams is and how it functions.

(3 marks)

Allows nose gear, which are designed to swivel, to establish straight wheel alignment, when LDGs are retracted in bay, and when aircraft is landing.

Centering is accomplished by centering cams:

- Cam rings are mounted around the top of the piston and bottom of the outer cylinder.
- When extended, the cam faces mate and hold the wheel in centered position.

4.3. Describe how stress corrosion can occur and what can you do to prevent corrosion it from taking place.

(3 marks)

Stress corrosion is when a stress applied to a material, there are new cracks or additional crack growth, which expose new areas. These newly exposed areas will corrode if not protected.

Choice of corrosion resistance material or other relevant anti-corrosion methods

- 4.4. Describe what is a “negligible damage” and state ONE example of it. (3 marks)

Negligible damage is a damage that does not affect the airworthiness of the aircraft. This level of damage is within the allowable limits per aircraft manufacturer’s Structural Repair Manual and exist as it is.

Minor dents or scratches or other relevant insignificant damages

- 4.5. State ONE advantage of using stacked doublers for a sheet metal repair. (2 marks)

By reducing the thickness of the patch material required at the first row to support the initial load transmission, this allows some weight saving by stacking doublers of different thickness together.

- 4.6. State ONE reason why it is important for a composite repair to be cured with the proper ramp up, dwell/soak time and cool down (2 marks)

It is to ensure proper heat transfer through the full thickness of the repair and component and to prevent any heat damage to the composite.

SECTION C – Analytical Questions (40 marks)

ALL diagrams are NOT drawn to scale.

Question 5

A section of the flight control linkage with pinned joints is shown in Figure 5. The structure is to be designed against buckling. Buckling can only occur in the X-Y plane (i.e. plane of this paper). With the information given below, determine the maximum load, F , before buckling occurs.

(12 marks)

- Both members are circular steel bars with a diameter of 70 mm
- Modulus of elasticity, $E = 150 \text{ GPa}$
- Factor of Safety = 3
- Moment of inertia for a circular cross-section, $I = \frac{\pi r^4}{4}$
- Euler's critical buckling load, $P_{cr} = \frac{\pi^2 E I}{L_e^2}$

Table 5: End Conditions and Effective Length	
End Conditions	Effective Length, L_e
Pinned-Pinned	$L_e = L$
Fixed-Fixed	$L_e = L/2$
Fixed-Free	$L_e = 2L$
Fixed-Pinned	$L_e = 0.7L$

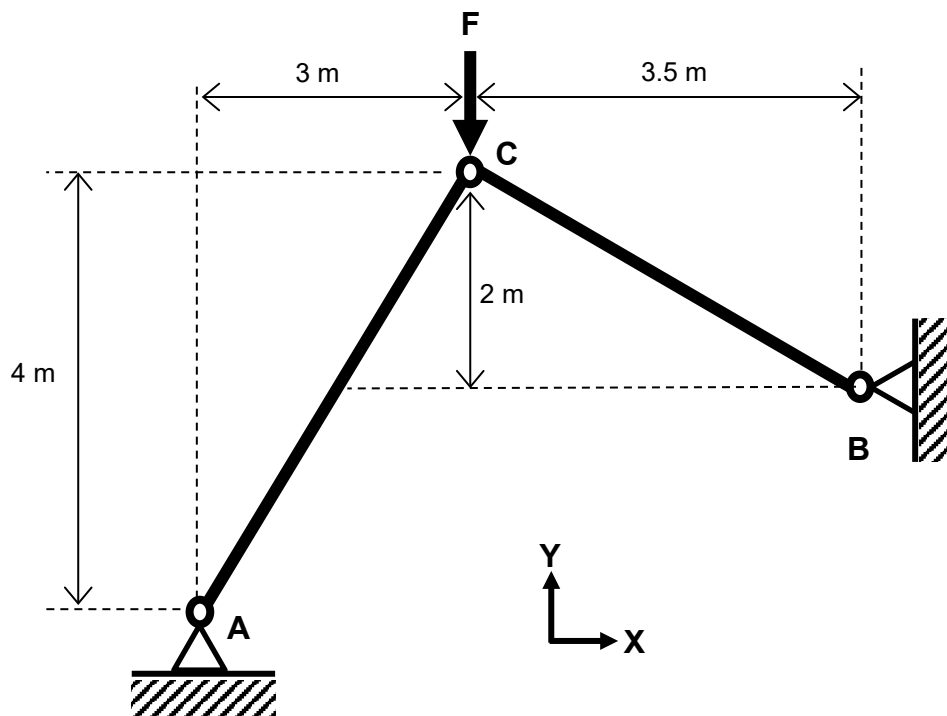


Figure 5

Question 6

An aircraft structural member, AB, with fixed-pinned end conditions is shown in Figure 6. The structure member is subjected to a distributed load as shown. Buckling can only occur in the X-Y plane (i.e. plane of this paper). With the information given, determine the maximum allowable value of distributed load, F, before buckling occurs. (12 marks)

- Member AB is a circular steel bar with a diameter of 60 mm
- Modulus of elasticity, $E = 150 \text{ GPa}$
- Factor of Safety = 3
- Moment of inertia for a circular cross-section, $I = \frac{\pi r^4}{4}$
- Euler's critical buckling load, $P_{cr} = \frac{\pi^2 E I}{L_e^2}$

Table 5: End Conditions and Effective Length	
End Conditions	Effective Length, L_e
Pinned-Pinned	$L_e = L$
Fixed-Fixed	$L_e = L/2$
Fixed-Free	$L_e = 2L$
Fixed-Pinned	$L_e = 0.7L$

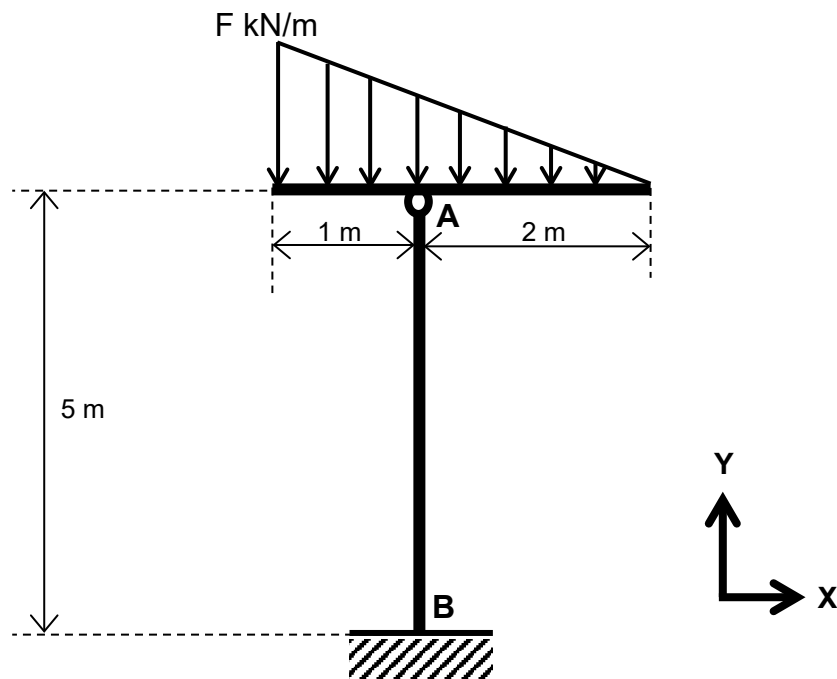


Figure 6

Question 7 (16 marks)

An overhanging section of a commuter category aircraft is illustrated in Figure 1. It is subjected to a distributed load point 1000 N/m over section AB and a point load of 5000 N at point C.

- 6.1 Using the method of integration with singularity functions, determine the deflection and slope equations that define this section of the spar.
- 6.2 If $EI = 10 \text{ MN.m}^2$, determine the slope at point B
- 6.3 If $EI = 10 \text{ MN.m}^2$, determine the deflection at point C.

Given,

$$EI \frac{d^2y}{dx^2} = M$$

where M = moment along any point along the span of the wing.

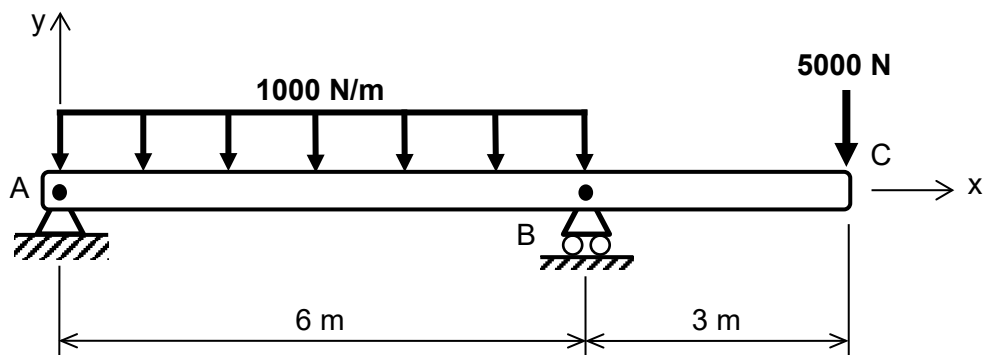


Figure 1

End of Paper